Campus Lost & Found

DevOps Final Capstone: one small three-tier app, taken from git push to a monitored Kubernetes cluster on AWS



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The application

M1

1

Frontend

React 18 + Vite on nginx-unprivileged; proxies /api to the backend

2

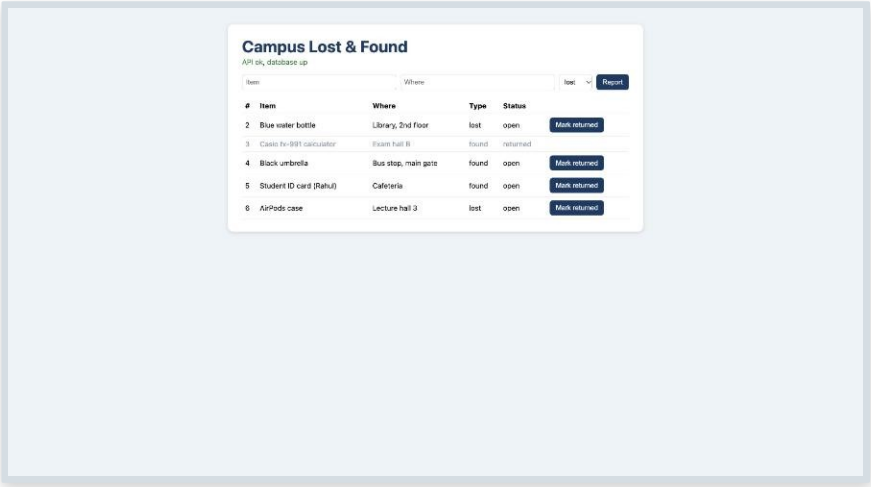
Backend

FastAPI, Python 3.12, psycopg 3, Alembic migrations, /metrics

3

Database

PostgreSQL 16, items table with check constraints

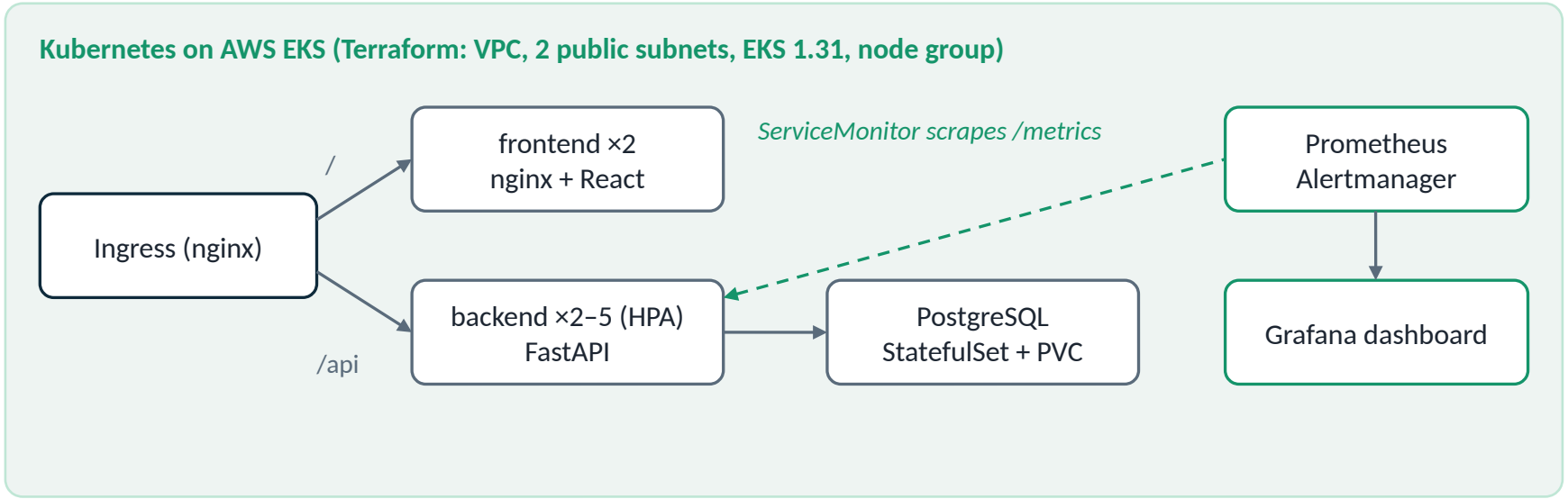


Running locally with docker compose at localhost:3000

CRUD on /api/items · /api/live · /api/health (503 if DB down) · /metrics

Architecture end to end





Tests and containers

M2–M4

9

pytest tests over 6 endpoints

87%

coverage, CI gate at 80%

3

Python versions in the test matrix

Docker

- Backend: python:3.12-slim, runs as uid 10001, HEALTHCHECK on /api/live
- Frontend: multi-stage, node:20 builds, nginx-unprivileged serves on 8080

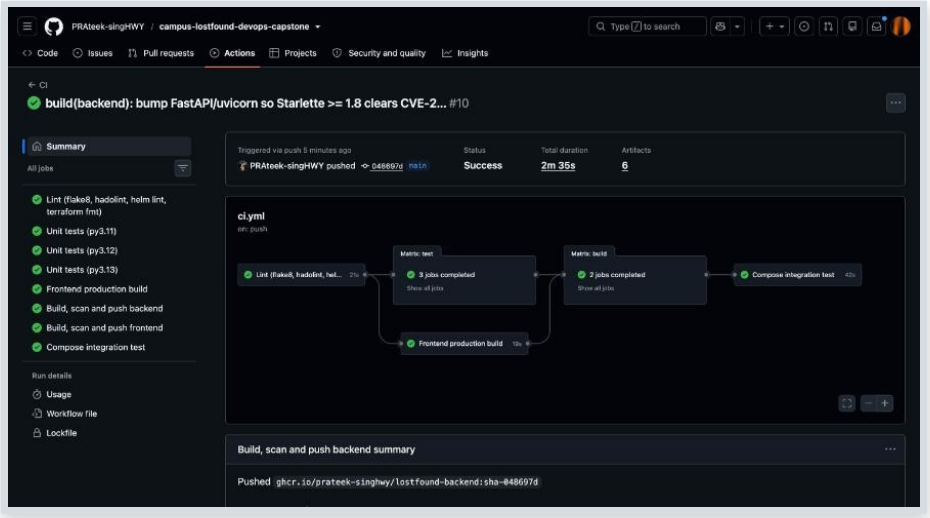
docker compose

- db → backend → frontend start order gated on healthchecks
- Two networks: the frontend cannot reach Postgres
- Named volume pgdata keeps data across restarts

CI pipeline on every push

M5

1. **lint**
   flake8 · hadolint · helm lint · terraform validate
2. **test**
   pytest on 3.11 / 3.12 / 3.13, 80% gate
3. **frontend-build**
   npm run build
4. **build**
   buildx → Trivy → push to GHCR
5. **integration**
   compose up from pushed images + smoke test



A green CI run: every job passed and both images were pushed

DevSecOps: the Trivy gate fired

M6

CRITICAL **CVE-2026-31789**

OpenSSL heap buffer overflow in libcrypto3 / libssl3 3.3.3

HIGH **CVE-2026-33630**

Use-after-free in c-ares

What the gate did

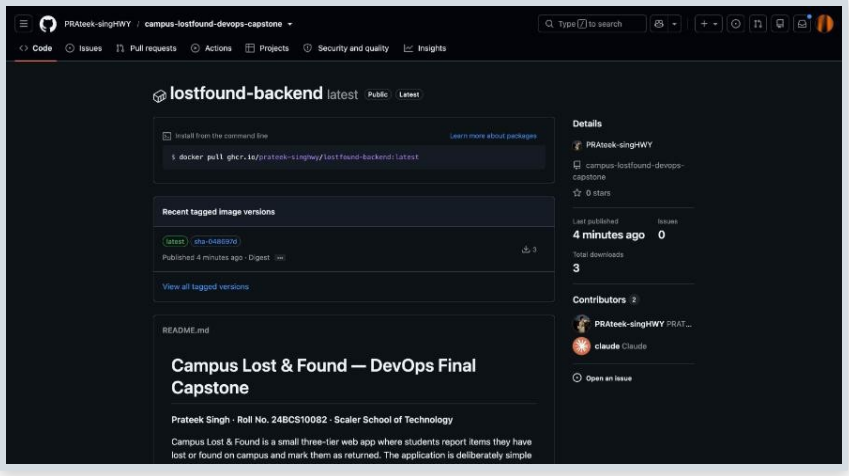
- severity HIGH,CRITICAL · exit-code 1 · ignore-unfixed
- Found both CVEs in nginx-unprivileged:1.27-alpine
- Blocked the build: no image was pushed

The fix

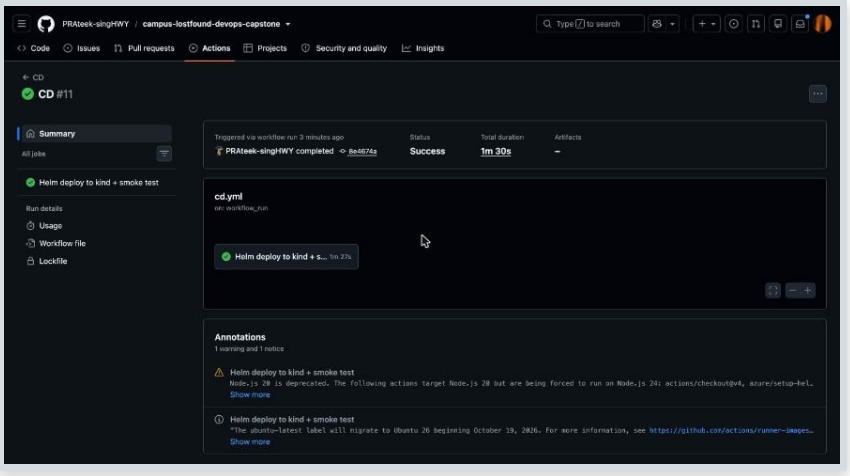
- Base image moved to 1.29-alpine + apk upgrade
- Backend got the matching apt-get upgrade
- Next run green; images pushed to GHCR

Delivery traced to a commit

M5



GHCR: images tagged sha-<commit>



CD: helm upgrade --install on kind, then smoke test

image.tag = sha-<commit> → any release can be reproduced, and helm rollback returns to a known image

Terraform on AWS

M7

26 AWS resources applied, then destroyed

2 t3.small workers (Free Tier)

1.31 EKS version, ap-southeast-2

`network.tf`	VPC 10.50.0.0/16, two public subnets tagged for ELB, IGW, routes
`eks.tf`	IAM roles, EKS control plane, node group 1–3, EBS CSI + metrics-server add-ons
`storage.tf`	Encrypted, versioned S3 bucket for pg_dump backups
`provider.tf`	use_localstack=true points every call at LocalStack for CI

Kubernetes and Helm on EKS

M8

Secret + ConfigMap

Postgres StatefulSet + PVC

Deployments ×2, maxUnavailable 0

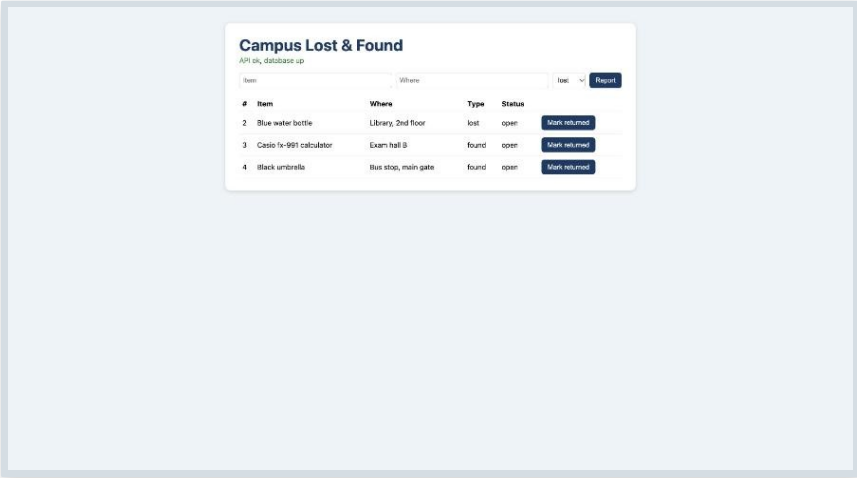
Startup / readiness / liveness probes

Ingress: /api → backend, / → frontend

HPA at 60% CPU

ServiceMonitor + PrometheusRule

values-dev / values-prod / values-eks



The app on EKS, served through an AWS load balancer

Observability

M9



Grafana: Campus Lost & Found dashboard

Metrics on /metrics

- http_requests_total
- http_request_duration_seconds
- lostfound_db_up · lostfound_items_open

5 alerts

- 5xx ratio high · p95 > 500 ms
- Database down
- Pod restarting · pod not ready

Real incidents, real fixes

Free Tier rejected t3.medium

Cause Node group failed to launch

Fix Switched workers to t3.small

EBS CSI crash-loop on EKS

Cause Controller could not reach IMDS for credentials

Fix Launch template with IMDSv2 hop limit 2

Backend restart on first install

Cause Postgres init outlasted the 15 × 2 s retry budget

Fix DB_RETRIES 30 + socket wait in entrypoint

Prometheus had no target

Cause ServiceMonitor label did not match the release

Fix Added the release label in the chart

From one commit to production

github.com/PRAteek-singHWY/campus-lostfound-devops-capstone

1. Edit and push a commit
2. CI: lint, test, Trivy, push sha tag
3. CD: Helm deploys that exact tag
4. helm rollback to any revision